

AI for Financial Practitioners

Lecture 3 · Part II

The Fortress and the AI Factory

Protect critical commitments. Experiment rapidly. Move authority only when evidence justifies it.

One institution — two complementary environments

Responsible innovation requires a protected operating core and a disciplined environment in which new capability can fail safely.



Fortress without Factory

Dependable, but increasingly stagnant.

Factory without Fortress

Inventive, but institutionally fragile.

The strategic design problem is the connection — not choosing one side

Cybersecurity is part of the architecture

As AI acquires tools, memory and persistent identity, every capability decision becomes a security and control decision.

Design for least privilege

- ▶ Separate read, write and send authority
- ▶ Distinct service identities and short-lived credentials
- ▶ Purpose-bound access to data and tools
- ▶ Segment experiments from production state

Design for intervention

- ▶ Monitor persistent authority over time
- ▶ Detect unusual access and action
- ▶ Suspend active authority quickly
- ▶ Reconstruct and reconcile what already happened

GOVERN

IDENTIFY

PROTECT

DETECT

RESPOND

RECOVER

A kill switch without logs is incomplete. Logs without stopping rights are incomplete.

The controlled membrane: from experiment to authority

Production authority should increase only as evidence, ownership, controls and recovery arrangements become stronger.



Evidence

Quality · edge cases ·
abstentions

Authority

Population · actions · limits

Operations

Owner · monitoring · rollback

A successful demo earns the right to be tested — not the right to act

AI economics: measure useful completion

A financial institution does not buy tokens; it funds a service that must deliver acceptable completed work.

$$C_{\text{case}} = C_{\text{data}} + C_{\text{compute}} + C_{\text{tools}} + C_{\text{review}} + C_{\text{exceptions}} + C_{\text{operations}}$$

Wrong denominator

Generated outputs

Rewards volume even when outputs require reconstruction.

Benefits

Separate cash savings, capacity, speed/coverage, decision quality and risk reduction.

Right denominator

Accepted useful results

Keeps economics connected to institutional purpose and acceptance standards.

Counterfactual

Compare against manual work, conventional software, narrower models — or stopping low-value work

Third-party dependence needs an exit architecture

The objective is not complete independence; it is informed dependence with visible concentration, substitutability and degradation paths.



Map concentration

Portfolio exposure may share one hidden failure domain.

Preserve portability

Interfaces, evaluations, records, procedures and performance history.

Test degradation

Know how the service behaves when a supplier changes or disappears.

A cheaper supplier is not cheaper if evidence, review, security or exit costs deteriorate.

Ownership turns a prototype into an institution

A production capability exists only when someone owns its purpose, operation, change, funding and withdrawal after launch.

Owner	Owns	Critical decision
Business	Purpose, users, benefits, acceptable outcomes	Expand or narrow the business scope
Service / operations	Reliability, incidents, lifecycle	Stop or degrade the workflow safely
Data	Authorized sources and uses	Permit or revoke changed data use
Engineering / model	Components and technical change	Release or revert a technical version
Risk / compliance / security	Challenge and control boundaries	Escalate material change or exposure
Senior governance	Cross-boundary trade-offs	Accept authority, risk and funding

Two questions reveal whether ownership is real: **Who can expand it?**
Who can stop it?

The board memorandum: eight questions

The board should fund a bounded institutional service — not an abstract commitment to “AI.”

1. Purpose What outcome and baseline?

2. Responsibility What does AI do; what remains human/deterministic?

3. Evidence What ordinary and severe cases support release?

4. Authority What exactly is delegated at launch?

5. Protection Which data, identities, records and services are exposed?

6. Economics Full cost per useful completion?

7. Dependencies Suppliers, fallback and exit path?

8. Ownership Who owns benefit, risk, operation, change and retirement?

Milestones should be evidence decisions — not calendar promises

Investment committee: fund the service, not the demo

A persuasive client-commentary demonstration is only the beginning of the investment decision.

FORTRESS TEST

- ▶ Authoritative portfolio and market data
- ▶ Client-data and communication boundaries
- ▶ Drafting separated from delivery
- ▶ Fallback when dependencies fail

ECONOMIC TEST

- ▶ Cost per acceptable commentary
- ▶ Review + exceptions + remediation
- ▶ Sensitivity to model price and review time

FACTORY TEST

- ▶ Unusual portfolios and stale prices
- ▶ Corporate actions / missing benchmarks
- ▶ Shadow-mode corrections categorized
- ▶ Negative results preserved

OWNERSHIP TEST

- ▶ Business owner + service owner
- ▶ Data / technical / risk ownership
- ▶ Vendor exit and reversion path

A rational decision may fund shadow operation while withholding client-delivery authority.

From models to investable institutional intelligence

The course ends where financial leadership begins: aligning capability, process, organization, authority, adaptation and investment into one coherent system.



The Fortress

Protects the commitments the institution cannot treat as experiments.

The AI Factory

Develops and evaluates the capabilities the institution does not yet possess.

Build intelligence that can be explained, challenged, improved, financed— and, when necessary, stopped.